FINVU INTEGRATION - WEB REDIRECT

FIU Integration using Web redirect

Version: 2.0

Confidential

This document provides details on the FIU integration with Finvu AA using Web redirect

Finvu Team
info@cookiejar.co.in

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Summary

In the AA ecosystem, FIUs are the consumers of customer data for providing various financial services. Hence majority of the customer journeys are initiated from the FIUs' application and services.
However, the consent approval for any data request from FIU needs to happen in the AA's domain. This could lead to broken journeys and bad user experience. This document provides information about APIs that FIUs can use to integrate Finvu Account Aggregator with their user journeys in order to provide seamless experience in the consent approval process.

Integration steps

FIU's application client running on user's device (e.g. browser or mobile app) can integrate with Finvu AA by using the web redirection.

AA interaction with the end customer has to be between the customer's device and AA server directly. Hence, the FIU client is expected to establish a connection from the user's device and not from FIU's servers to Finvu AA server.

Finvu AA Web redirection provides a secure, stateful and authenticated session to the end users in order for them to login to their AA account using OTP based authentication and provide consent. OTP based authentication will ensure that users need not use their password to authenticate to AA, which means the users' credentials cannot be stored by an FIU.

The username and password received from Finvu Team will be used for getting the token.

Below steps to integrate for raising a consent and fetching data.

Server to server REST API calls

Step 1) Get a token to call the REST API.

Endpoint

https://<fiuname>.fiu.finfactor.in/finsense/API/V1/User/Login

Method: POST

Request Body

```
{
    "header": {
        "rid": "42c06b9f-cc5b-4a53-9119-9ca9d8e9acdb",
        "ts":  "2019-07-15T11:03:44.427+0000",
        "channelId": "finsense"
    },
    "body":{
            "userId": "<user name>",
            "password": "<password>"
    }
}
```

Response


```
{
    "header": {
        "rid": "42c06b9f-cc5b-4a53-9119-9ca9d8e9acdb",
        "ts": "2020-02-21T12:22:56.924+0000",
        "channelId": "finsense"
    },
    "body": {
        "token":
"eyJraWQiOiJyc2ExIiwiYWxnIjoiUlMyNTYifQ.eyJpc3MiOiJjb29raWVqYXIiLCJhdWQiOiJjb29raWVqYX
IiLCJleHAiOjE1ODIyODk1NzYsImp0aSI6InRXa0hJYjh4SnVpenFGRlJDQ2Jsb3ciLCJpYXQiOjE1ODIyODc3
NzYsIm5iZiI6MTU4MjI4Nzc3Niwic3ViIjoibWFuaXNocyIsInJvbCI6ImFkbWluIn0.InYHWtddwr3cD5JXn1
nMiaNAK-
dzUKKaaqubc4cy504zSHeHn4LrBByxg6baXgmDnkVl9MNTKtCNHnJDkgnrmc9R3lA1WUHEdCZtdQjYJtGYlqjX
lxSpABQp05_GBFg2FALdE4tcy3aGBK0J-
ByKaMuAX5FHh1spldEqaiZrSLYK6Buty97Eo9SXqLXFf6yLxRsCH8KwusUMrz4lSSs6_Jmgf3XnfUGhEVtfgDC
zIphZ9apB1GP3F8QV2_D6tZ61VF6gfqihARhdvDyTWt0stDrZNPeGWJI1rkUptc7xnytaHYZXFnLxzGx1HQCVL
EkwqORjCdM4VOqWB-Z6SBDdFQ"
    }
}
```


NOTE : This token is to be used in the SERVER to SERVER API calls and passed in the HTTP header ‘Authorization’ with value Bearer: <token>

The Token is valid for 24 hrs can be refreshed once per day.

Step 2) Initiate a REST API request to the server for raising a consent request

This returns encrypted parameters for redirection or SDK

Endpoint

`https://<fiuname>.fiu.finfactor.in/finsense/API/V1/ConsentRequestPlus`

HTTP header

Set HTTP header 'Authorization' with value Bearer: <token received in Step 1)>

Method: POST

Request Body


```
{
    "header": {
        "ts": "2023-02-08T12:24:01.890+0530",
        "channelId": "finsense",
        "rid": "1100000000289243"
    },
    "body": {
        "custId": "9860823869@finvu",
        "consentDescription": "Personal finance management",
        "templateName": "FINVUDEMO_TESTING",
        "userSessionId": "sessionid123",
        "redirectUrl": "https://google.co.in",
        "fip" : [""],
        "ConsentDetails": {
        },
        "aaId": "cookiejar-aa@finvu.in"
    }
}
```


The Template Name can be the template that is created and is configurable.
The userSessionId is optional can be any alpha numeric string which can be used by the FIU for tracking purpose.

Response


```
{
    "header": {
        "rid": "1100000000289243",
        "ts": "2023-06-22T04:38:14.658+00:00",
        "channelId": "finsense"
    },
    "body": {
        "encryptedRequest": "Dwyp4YhGR07aDbA80mZ30TEtRe_NaPH-
2IUkEYsFgC8Dp5CRpSwaC8FdOLdvrATPaTIkO7B8uw1Sr66R2Go9p2Iajqovbg7sKPxC7Jkz3JRn-
z40xe0xtuVjJ1Zn9GZPUfgr4fadghQl7HwUt8WEDB2oPYaLDlMvY29N264gWLx7jVjBYKwFgTVzQUEQLl2UZIDnzBulAV
8TZHKHj_uP9krOqW6kAhe0EEDSgfFU_Cc=",
        "requestDate": "220620230438694",
        "encryptedFiuId": "VFtFdlZYU11RREFZT1ZTUw==",
        "ConsentHandle": "fe7dd9b4-406e-413e-b6b4-8be730a94c41",
```

```
        "url": "https://reactjssdk.finvu.in/?ecreq=Dwyp4YhGR07aDbA80mZ30TEtRe_NaPH-
2IUkEYsFgC8Dp5CRpSwaC8FdOLdvrATPaTIkO7B8uw1Sr66R2Go9p2Iajqovbg7sKPxC7Jkz3JRn-
z40xe0xtuVjJ1Zn9GZPUfgr4fadghQl7HwUt8WEDB2oPYaLDlMvY29N264gWLx7jVjBYKwFgTVzQUEQLl2UZIDnzBulAV
8TZHKHj_uP9krOqW6kAhe0EEDSgfFU_Cc=&reqdate=220620230438694&fi=VFtFdlZYU11RREFZT1ZTUw==&fip="
    }
}
```


Note: Ensure that you store the consentHandle id

Step 3) Redirect the FIU frontend to the Finvu AA fronted

Construct the URL as below and redirect

https://reactjssdk.finvu.in?ecreq=XXXXXX&reqdate=XXXXXXX&fi=XXXXXXX

ecreq -> encryptedRequest (response from Step 2))

reqdate -> requestDate (response from Step 2))

rfi -> encryptedFiuId (response from Step 2))

The user will carry out the actions on the Finvu AA frontend to link and approve the consent request. Once the consent is accepted/denied, Finvu AA will redirect the user back to FIU

Step 4) Decrypt the response from the Finvu AA frontend

Endpoint
https://<fiuname>.fiu.finfactor.in/finsense/API/V1/Webview/Decrypt

HTTP header

Set HTTP header 'Authorization' with value Bearer: <token received in Step 1)>

Method: POST

JSON request object key mapping:-

Object name	**key name**	**key type**	**key description**
header			
	rid	UUID	Request Id in UUID format
	ts	timestamp	Timestamp of the request. Format is yyyy-mm-dd'T'hh:mm:ss.misss'+'<timezone offset> time in 24 hour format, last 3 digit of time are milliseconds.
	channelId	String	Hardcode to "finsense"
body			
	encryptedRequest	String	To be taken from the 'ecres' field passed in GET http URL mentioned in section 'Response (AA calling FIU)'
	RequestDate	timestamp	To be taken from the 'resdate' field passed in GET http URL mentioned in section 'Response (AA calling FIU)'
	encryptedFiuId	String	Encrypted FIU id of the FIU. To be taken from the 'fi' field passed in GET http url mentioned in section ''Response(AA calling FIU)'
	aaId	String	hardcode to "cookiejar-aa@finvu.in"

Sample JSON request body:-

{

"header": {
"rid": "f871126b-dd3f-4bb4-be8c-b1911b57a893",
"ts": "2020-01-13T12:58:33.501+0000",
"channelId": "finsense"
},

"body": {
"encryptedRequest":
"sSrBG8oYIveXbgEwE2GJ5stCth2saXs_Jc13yiDBvwdKfQnzd7VroZaakE2i4tP4D528OSgha9SxJ-gylsns6mE3gHiwJ8SsyWGU1Zy02VBlY2aO3zOZBNIJWeI4N56GqhGd4xXAfOCcmtCqzuLshftDtotm4kAwRqy6ukRwzJ21gnlhBeXXP--RY9lB8xANVPrMUZFsGEJNF3f61PF-I9IyqOnv69R8rucsxpvp8neILjQkecMd1FwNLc4W4Oob",
"requestDate": "040120212014647",
"encryptedFiuId": "040120212014647",
"aaId": " cookiejar-aa@finvu.in "
}
}

JSON response object key mapping:-

Object name	**key name**	**key type**	**key description**
header			
	rid	UUID	Request Id in UUID format
	ts	timestamp	Timestamp of the request. Format is yyyy-mm-dd'T'hh:mm:ss.misss'+'<timezone offset> time in 24 hour format, last 3 digit of time are milliseconds.
	channelId	String	Hardcode to "finsense"
body			
	txnId	UUID String	Original transaction id of the request
	sessionId	String	Original session id of the request
	srcRef	String	Original consent handle Id
	userId	String	Original AA user id of the finvu user
	redirectUrl	String	Original FIU domain url to which redirection to be done
	FiuId	String	Original registered Id of the FIU
	status	String	"S" in case of success, "F" in case of failure.
	errorCode	String	"0" in case of success, "1" in case of failure

Sample JSON response body

{
"header": {
"rid": "f871126b-dd3f-4bb4-be8c-b1911b57a893",
"ts": "2021-01-04T14:45:57.866+0000",
"channelId": "finsense"

},
"body": {
"txnId": "acf5dce8-cbf7-4f37-b03a-4a018e2dd1e4",
"sessionId": "asd12345",
"srcRef": "09c16861-e29b-42d1-b5a1-48716c05edd5",
"userId": " aauser@finvu ",
"redirectUrl": "<fiu redirect url>",
"fiuId": null,
"status": null,
"errorCode": null
}
}

Step 5) Check consent status

Once user has accepted the consent in Finvu after web redirect

Server can check status of consent and get the consent id

Endpoint
https://<fiuname>.fiu.finfactor.in/finsense/API/V1/<consentHandleId>/<custId>

HTTP header

Set HTTP header 'Authorization' with value Bearer: <token received in Step 1)>

consentHandleId from Step 1
custId : mobileno@finvu

Method : GET

Response :

Returns the consentId:null and consentStatus: "PENDING" in the body. When the user accepts the consent on the Finvu Webview or Finvu Mobile app, consent status of 'ACCEPTED' will be returned, and a valid consent will be present in the consentId field.


```
{
    "header": {
        "rid": "5c9d403d-d9c1-4177-b27e-3aca5a660938",
        "ts": "2019-07-15T11:03:44.427+0000",
        "channelId":  "finsense"
    },
    "body": {
        "consentStatus": "PENDING",
        "consentId": null
    }
}
```


Response :

Returns the consentId and consentStatus: “ACCEPTED” in the body. When we say accept from mobile app.


```
{
    "header": {
        "rid": "03ffe53c-89e4-4197-ada7-aafac32cf14a",
        "ts": "2019-07-15T11:07:29.696+0000",
        "channelId":  "finsense"
    },
    "body": {
        "consentStatus": "ACCEPTED",
        "consentId":  "117e8942-1605-4b75-b2db-4b8c953abeae"
    }
}
```


Step 6) Trigger a data request

Endpoint

https://<fiuname .fiu.finfactor.in/finsense/API/V1/FIRequest

HTTP header

Set HTTP header ‘Authorization’ with value Bearer: <token received in Step 1)>

Method : POST

Request:


```
{
          "header": {
                    "rid": "42c06b9f-cc5b-4a53-9119-9ca9d8e9acdb",
                    "ts":  "2019-07-15T11:03:44.427+0000",
                     "channelId":  "finsense"
          },
          "body":{

                    "custId": "mobileno@finvu",
                    "consentId": "117e8942-1605-4b75-b2db-4b8c953abeae",
                    "consentHandleId":"6fa1b74b-62b8-48dd-be2d-696b978aaf8f",
                    "dateTimeRangeFrom": "2018-09-30T08:10:45.006+0000",
                    "dateTimeRangeTo": "2019-09-30T08:10:45.006+0000"
          }
}
```


Response:


```
{
    "header": {
        "rid": "41f44ca2-4b9a-4310-b6a0-eaaa1349bdd6",
        "ts": "2019-07-15T11:08:21.722+0000",
        "channelId":  "finsense"
    },
    "body": {
        "ver": "1.1.2",
        "timestamp": "2019-07-15T11:08:21.404+0000",
        "txnid": "eeddca07-bccb-4808-8b4a-573322223566",
        "consentId": "117e8942-1605-4b75-b2db-4b8c953abeae",
        "sessionId": "eeb30580-3740-42c4-887b-3f08b929be58",
        "consentHandleId": null
    }
}
```

Step 7) Check the Data request status

Endpoint
https://<fiuname>.fiu.finfactor.in/finsense/API/V1/<consentHandleId>/<custId/<consentId>/<sessionId>

HTTP header
Set HTTP header 'Authorization' with value Bearer: <token received in Step 1)>

consentHandleId from Step 1
custId : mobileno@finvu
consentId from Step 5) when Consent has ACCEPTED status
sessiondId from Step 6) after initiation a data request

Method : GET

Response:

```
{
    "header": {
        "rid": "7e61e4d8-f9db-488b-848e-331db1c90f16",
        "ts": "2019-07-15T11:10:32.987+0000",
        "channelId":  "finsense"
    },
    "body": {
        "fiRequestStatus": "PENDING"
    }
}
```

When data is ready return response

Response:

```
{
    "header": {
        "rid": "7e61e4d8-f9db-488b-848e-331db1c90f16",
        "ts": "2019-07-15T11:10:32.987+0000",
        "channelId":  "finsense"
    },
    "body": {
        "fiRequestStatus": "READY"
    }
}
```

Step 8) Fetch the data

Endpoint
https://<yourfiuname>.fiu.finfactor.in/finsense/API/V1/FIDataFetch/<custId>/<consentId>/<sessionId>

HTTP header
Set HTTP header 'Authorization' with value Bearer: <token received in Step 1)>

custId : mobileno@finvu
consentId from Step 5) when Consent has ACCEPTED status

sessiondId from Step 6) after initiation a data request

Method : GET
Response


```
{
    "header": {
        "rid": "bda3cc3e-1c6e-4994-b8dd-b1f3d671d92a",
        "ts": "2021-08-20T04:30:53.479+0000",
        "channelId": null
    },
    "body": [
        {
            "fiObjects": [
                {
                    "Profile": {
                        "Holders": {
                            "Holder": {
                                "name": "Finvu",
                                "email": "finvu@fourcloud.in",
                                "dob": "1985-09-24",
                                "mobile": "8421068174",
                                "ckycCompliance": true,
                                "pan": "CGDMP1711L",
                                "nominee": "NOT-REGISTERED"
                            },
                            "type": "JOINT"
                        }
                    },
                    "Summary": {
                        "type": "SAVINGS",
                        "branch": "Pune",
                        "facility": "CC",
                        "ifscCode": "BARB0SHIPOO",
                        "micrCode": "411012029",
                        "openingDate": "2018-04-01",
                        "currentODLimit": "12000",
                        "drawingLimit": "30000",
                        "Pending": {
                            "amount": 20.0
                        },
                        "currentBalance": "100000",
                        "currency": "INR",
                        "exchgeRate": "2000",
                        "balanceDateTime": "2004-04-12T18:20:00Z",
                        "status": "ACTIVE"
                    },
                    "Transactions": {
                        "Transaction": [
                            {
                                "type": "CREDIT",
                                "mode": "ATM",
                                "amount": 32850.0,
                                "currentBalance": "70250",
                                "transactionTimestamp": "2020-01-01T18:20:00Z",
                                "valueDate": "2018-02-04",
                                "txnId": "0ff0129a-9ae0-4933-8f1b-79ff4c45966c",
                                "narration": "Rent for 2019-20:-Locker No:33",
                                "reference": "REF8421068"
                            },
                            {
                                "type": "DEBIT",
                                "mode": "ATM",
                                "amount": 32850.0,
                                "currentBalance": "70250",
                                "transactionTimestamp": "2020-04-04T18:20:00Z",
                                "valueDate": "2019-01-03",
                                "txnId":  "eb195441-438a-48fa-b625-347234a0739b",
                                "narration": "INF/INFT/0229857474/SALARY Mar 19",
```

```
                                    "reference": "REF8421068"
                                },
                                {
                                    "type": "CREDIT",
                                    "mode": "ATM",
                                    "amount": 30000.0,
                                    "currentBalance": "50000",
                                    "transactionTimestamp": "2020-07-04T18:20:00Z",
                                    "valueDate": "2020-03-04",
                                    "txnId": "0ff0129a-9ae0-4933-8f1b-79ff4c45966c",
                                    "narration":
"MMT/IMPS/74784747/Family/CBIN247474",
                                    "reference": "REF8421068"
                                },
                                {
                                    "type": "CREDIT",
                                    "mode": "ATM",
                                    "amount": 30000.0,
                                    "currentBalance": "10000",
                                    "transactionTimestamp": "2020-07-15T18:20:00Z",
                                    "valueDate": "2020-05-10",
                                    "txnId": "0ff0129a-9ae0-4933-8f1b-79ff4c45966c",
                                    "narration": "NEFT/U000000633712182/BOB
SAMADHAN/PENSION MAR 206",
                                    "reference": "REF8421068"
                                }
                            ],
                            "startDate": "2019-08-20",
                            "endDate": "2022-01-20"
                        },
                        "type": "deposit",
                        "maskedAccNumber": "BOB8421068",
                        "linkedAccRef": "REF8421068",
                        "version": "1.1"
                    }
                ],
                "fipId": "BARB0KIMXXX",
                "fipName": "Finvu Bank Ltd.",
                "custId": "finvudemo@finvu",
                "consentId": "805d8536-bac5-498f-a863-da09feb842fa",
                "sessionId": "3443cb71-0a6d-417b-987e-6f52b955751f",
                "fiAccountInfo": [
                    {
                        "accountRefNo": "REF8421068",
                        "linkRefNo":  "731733ac-9bc5-486a-86f2-3d0ebd73c7e6"
                    }
                ]
            }
        ]
    }
```


By Changing the Header （Accept: application/xml） you will be able to download the data in different format- json/pdf. PDF will be in binary format.

```
byte[] bArray= "response of API"
OutputStream out = new FileOutputStream("out.pdf");
out.write(bArray);
out.close();
```

Callback Notification APIs (Optional if required)

Consent Notification

```
{{Notification endpoint}}/Consent/Notification
```

This api is called by Finsense when the consent is approved by the user and when Finsense receives notification from AA. Finsense will then fetch the consent from AA and notify the channel/application that the consent has been approved/rejected (or any other consent life cycle event)

Request

```
{
        consentHandle: "590b771e-ca09-4673-8291-c769db2f5949",
        status:
"ACTIVE|PENDING|REVOKED|PAUSED|REJECTED|EXPIRED|FAILED",
        consentId: "9f0f6d31-1316-44dd-8d6a-0b9bcac0d807"
 }
```

In response, the channel/application should response with http response code 200, and body is not expected from the channel.

Data Notification

```
{{Notification endpoint}}/FI/Notification
```

This api is called by Finsense when the data is received by Finsense from AA.

```
{
        consentHandle: "590b771e-ca09-4673-8291-c769db2f5949",
        dataSessionId: "f903f1de-d47b-4ab6-bde7-c666add4a3e8"
        status: "READY|FAILED"
}
```

In response, the channel/application should response with http response code 200, and body is not expected from the channel.

After receiving the `/FI/Notification`, the channel/application is expected to run the `/FIDataFetch` api to retrieve data from Finsense.

Data Fetch

```
/FIDataFetch/{consentHandleId}/{sessionId}
```

This is a simplified API that can be used to fetch data once the data ready notification is received by the channel/application. This is a GET request.